

CE2200 – Statics
Fall 2025
Final Project:
____/100pts

Honor code:

All the work submitted here is our own work on this project this semester. We haven't used any previous work from this or other courses. All the text produced in this report is our text and wasn't automatically generated using AI. We are allowed to use AI to enhance our writing but not to generate original text. All work that we didn't produce is properly cited. We haven't received unauthorized help in crafting this report.

Signatures of authors:

Mateo Jaime Tecum Jacinto

Auris Vega

Contents

Table of Figures:	3
Table of Tables:	3
Step 1: Federal Highway Act research (10 pts)	4
Step 2: Preliminary Truss Design (30 pts).....	6
Step 3: Building the truss prototype (10 pts)	8
Step 4: Calculating for failure (10pts)	10
Step 5: Comparing calculations with lab results (10pts)	12
Step 6: Designing the Community Bridge (10pts)	14
Step 7: Checking for Stability on Support (5pts).....	19
Step 8: Conclusion and letter to the community (10pts)	20
Quality and clarity of report: (5pts)	Error! Bookmark not defined.

Table of Figures:

Figure 1: Truss	6
Figure 2: Example of a load combination.....	14
Figure 3: Truss with Horizontal Load.....	19
Figure 4: Actual Truss	
Figure 5: <i>Measurements listed to actual truss in class</i>	
Figure 6: <i>post Test, Truss has no deformities, except for broken weld on edge</i>	
Figure 7: Example of a load combination	
Figure 8. <i>Global free-body diagram of the bridge truss under Load</i>	
Figure 9. <i>Global free-body diagram of the bridge diagram of the bridge truss</i>	
Figure 10. <i>Global free-body diagram of the bridge diagram of the bridge truss</i>	
Figure 11. <i>Joint A free-body diagram illustrating the forces in members AB and AH and the vertical support reaction at A</i>	
Figure 12. <i>Joint C free-body diagram showing the axial forces in members of BC, CD, CG, and HC used in the method-of-joints analysis.</i>	
Figure 13: <i>Truss with Horizontal Load</i>	
Figure 14. <i>Work shown for parts A, B, C</i>	

Table of Tables:

Table 1: Forces in Figure 1: Truss	
Table 2: Coordinates of the Nodes (joints) in the designed truss...	
Table 2: Members Node length and force of the designed truss....	
Table 3: External forces of the truss.....	
Table 4: constants known for A-36 steel alloy with pin.....	
Table 5: List of yielding members.....	
Table 6: buckling members.....	

Step 1: Federal Highway Act research (10 pts)

You should aim for 500-750 words. Make sure you cite all your external sources using the References tab above (you may pick a style you are comfortable using) and use reliable sources. The bibliography for this section will be included in the bibliography of the whole project at the end of the report.

The Federal-Aid Highway Act of 1956 launched the Interstate Highway System with a set of national objectives grounded in transportation safety, economic efficiency, and national defense. President Dwight D. Eisenhower and federal planners argued that modern highways were essential for reducing the era's high roadway death toll, alleviating congestion, improving commercial transport, and enabling rapid mobilization for civil defense during the Cold War (Weingroff, 2025). The program served as a large work project that was envisioned to help the economy through more transportation that would lead to large-scale employment. The system itself was enacted through a lot of bureaucracy and arguing between states and the government on funding. Eventually, the federal government provided 90 percent of the funding and set strict standards for what these highways were supposed to look like. The state highway departments and local officials oversaw setting both the route and construction for the interstates as a result. This meant that through this large pool of funding from the government, states were strongly encouraged to route interstates to help build the economy. As a result, states routed these interstates through urban areas, as it was seen as cheaper in terms of land acquisition (Mohl, 2004). City officials and state departments were determined to grow their cities with these interstates, and so they had supported both the engineering and the cost efficiency of where the interstate would go. Although the construction was cost efficient and helped the city, the construction of the interstate system had a devastating and unfortunate impact on underrepresented minorities, also known as URM communities, particularly the Black neighborhoods. In Nashville alone, the routing of Interstate 40 (I-40 for short), had cut through Jefferson Street, which was a thriving business district, leaving it to be getting rid of the Black culture there was. Former Councilmember Erica Gilmore admits that the routing had erased the hub of Black commerce and culture, divided the community and destroyed homes in the process as well (Gilmore, 2021). This incident is only one example of a pattern that emerged from the Interstate Highway Program. Other cities that had erased minority

neighborhoods included I-80 and I-10 in Syracuse and New Orleans. The consequences of this routing today have only led to many racial and economic disparities. Neighborhoods in proximity or cut off by interstates must also deal with reduced property values and pollution, on top of less economic opportunities. In Nashville, I-40 has resulted in an unsafe pedestrian environment within the city, and if we wish to repair the damage, initiative must be taken to rebuild the communities that were lost. Some examples of these movements include the Jefferson Street Multimodal Cap and Connector project which created a lot of public space near I-40.

There are some other similar projects that are fighting the same issue in other cities, which only emphasize the idea that we can in fact make an impact and fix our shortcomings in the past. In Dallas, the Southern Gateway park caps a portion of I-35E in the Oak Cliff neighborhood, which reunited the community with lawns and walkways, and even a multipurpose building. This all happens while incorporating feedback from over 500 residents (Nemo, 2023). In Denver, a four-acre cover park over I-70 aims to reconnect the Globeville Elyria-Swansea neighborhood, which is a predominantly Latino area, though it has raised concerns about gentrification and displacement (Hammon, 2023). There are also works in Los Angeles, California, where the proposed Belveder Park Highway Cap over the 60 Freeway would reconnect the north and south sections of the park and improve pedestrian and bike access (FHWA, n.d.). All these cities, and their dedication to fighting the difficulties that interstates accidentally made should drive us to address the impacts of interstate even more, especially considering the fact we need to be fighting for both of us and for the underrepresented communities.

Step 2: Preliminary Truss Design (30 pts)

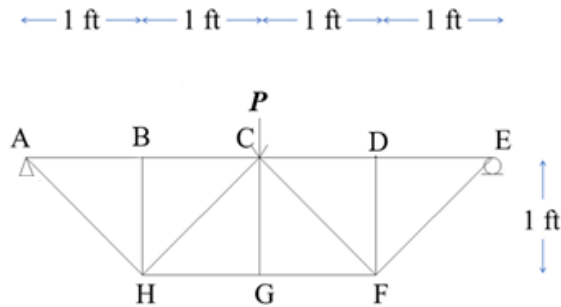


Figure 1: Truss

A) $\sum F_y = 0 = A_y - F_{AH} \cos(45^\circ) = 0$
 $A_y = F_{AH} \cos 45^\circ$
 $\sum F_x = 0 = F_{AB} - F_{AH} \sin 45^\circ = 0$
 $F_{AB} = F_{AH} \sin 45^\circ$
 $F_{AB} = 175 \text{ lb}$
 $F_{AH} = 175 \sqrt{2} = 247.49$

B) $\sum F_y = 0 = F_{BH} = 0$
 $\sum F_x = 0 = F_{BC} = F_{AB} = 175 \text{ lb}$

C) $\sum F_x = 0 = F_{BC} - F_{CD} + F_{CH} \sin 45^\circ + F_{CF} \sin 45^\circ = 0$
 $F_{CD} = 175 \text{ lb}$
 $F_{CH} = 175 \text{ lb}$
 $F_{CF} = 247.49 \text{ lb}$

D) $\sum F_x = 0 = F_{DE} = 0$
 $\sum F_y = 0 = F_{DF} = 0$
 $F_{DE} = 0$
 $F_{DF} = 0$

E) $\sum F_y = 0 = E_y - F_{EF} \cos 45^\circ = 0$
 $E_y = F_{EF} \cos 45^\circ$
 $F_{EF} = 175 \text{ lb}$
 $F_{EH} = 175 \text{ lb}$
 $F_{CF} = 247.49 \text{ lb}$

F) $\sum F_y = 0 = F_{GF} \cos 45^\circ + F_{CF} \cos 45^\circ = 0$
 $F_{GF} = -247.49 \text{ lb}$
 $F_{CF} = 247.49 \text{ lb}$

Component

$F_{BH} = 0$
 $F_{CH} = 175 \text{ lb (C)}$
 $F_{AH} = 247.49 \text{ lb (T)}$
 $F_{DE} = 175 \text{ lb (C)}$
 $F_{EF} = 247.49 \text{ lb (C)}$
 $F_{CD} = 175 \text{ lb (C)}$
 $F_{BC} = 175 \text{ lb (C)}$
 $F_{CH} = 175 \text{ lb (C)}$
 $F_{CF} = 247.49 \text{ lb (C)}$
 $F_{GF} = -247.49 \text{ lb (C)}$
 $F_{GF} = 247.49 \text{ lb (C)}$

Figure 2: Method of joints calculations for internal forces

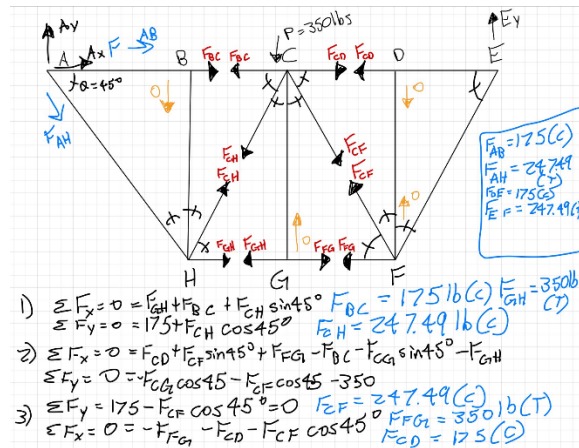


Figure 3: Method of sections calculations and free body diagram

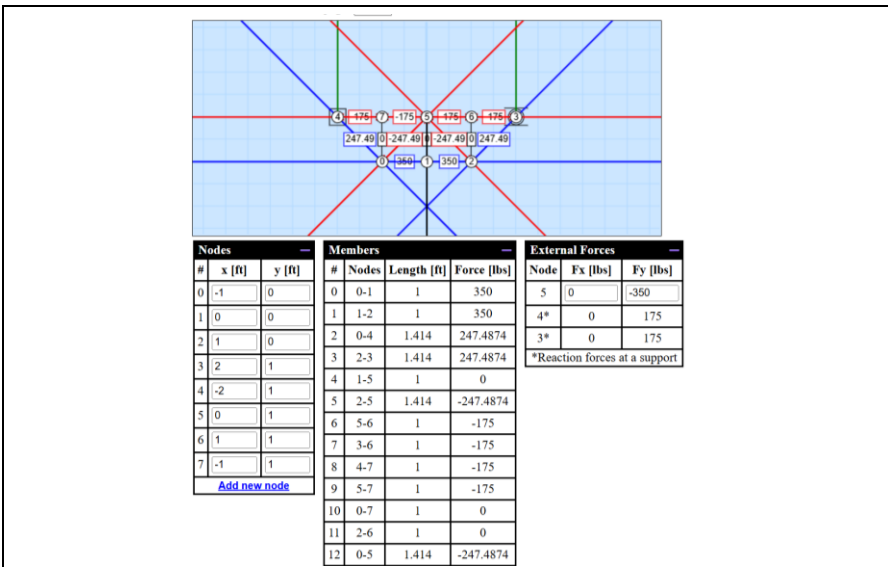


Figure 4: Graph showing the solution of the truss (ft, lbs for units)

Tables 1-3: Node. Members, and external forces

The results from the software both support my numbers (other than the slight rounding issue) and even support the directions that I set my reactions and

forces (either compression or Tensions), which will allow us to see which members can buckle and which ones can possibly yield.

Step 3: Building the truss prototype (10 pts)

Our team started by getting familiar with the tools when we first went to weld during our statics class. This involved practicing and understanding how welding works and then welding two solid plates together for our first “lab” session. The following week, we moved onto more solid plates together, until we finally moved onto fully tacking and welding (or tack-weld) our members for our truss section. We had been provided with smaller but long steel plates that were then tacked and welded to serve as the members in our truss. This typically included two tacks on the edge of the long steel plate that was positioned perpendicular to the smaller steel plate, that also included a slot for the large steel plate to fit in. After tacking we flipped the side over and then began doing a full weld on the other side. While completing this step, it was necessary for our TAs to ensure our welds were good. To determine whether a weld and tack were suitable for our truss, we usually had to look for the heat signature that was left on the plate after welding, a good 45 degree welding angle on the two plates, and also to physically check, we would apply a decent amount of force on the welded plates to see if the weld would break. If this happened, we would need to try again, which resulted in a harder tack because of the already existing tack and weld from the previous attempt.

After welding our members, we then moved onto the main frame and tackled the assembly of the members to a top and bottom steel plate. Not only did it also include tack-welds, but alignment was key. Our job was to align the members together between the plates and have the clamps stand nicely (and hopefully not move when welding), and then do the typical tack-weld. The protocol for how we went about this portion of the design and creation was a little rushed, and could be improved, as we did in fact have a misalignment that led to a mediocre weld that stuck, but we did not know how strong the weld was until the Test Day. This was our biggest challenge, on top of getting the tack-weld correct the first time in order to get the best result of our members. Then, in the next session, our team collaborated with two other teams to combine our members and create full truss. After alignments, tack-welds, and the screws that we needed to include during this session, our truss turned out to look prepared for Test Day.

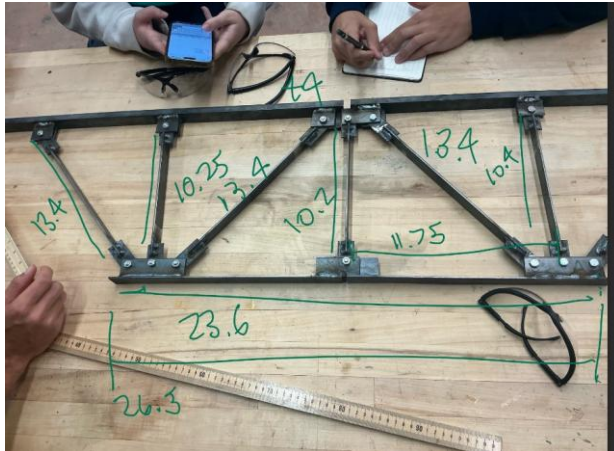


Figure 5: Measurements listed to actual truss in class

It painfully became clear during the results that the assembly was done incorrectly. When applying the load while testing, our truss had only been able to withstand a couple hundred pounds of force, which was substantially lower than the rest of the groups. We observed while measuring and looking at our truss, there was not too much stress on the actual members, but there was indeed a weld that was not making full contact with its partner when welded, which resulted in a very weak connection as seen in the figure below:



Figure 6: post Test, Truss has no deformities, except for broken weld on edge

With a max load of 215lbs, our truss could not be observed in terms of its stress on the members. Rather than having the desired failures mentioned later in the report, our team had a failure that was abnormal and only occurred due to a rush in welding and misaligning the members together. As you would see later, our theoretical calculations had predicted a higher load capacity, but the reality of having a weld fail first was brutal and led to a weak reported “strength of the truss”.

Step 4: Calculating for failure (10pts)

To assess the safety of our truss design, our failure modes that we needed to analyze included the yielding in tension and the buckling (in compression), for this the yielding failure just means that the material is failure to uphold the stress that is caused on the tension member. In the equation given, this can be seen as the material's yield strength being less than the calculated stress given by the equation $\sigma = \frac{T}{A}$. When this occurs, the member can physically be seen as deformed, and in our case since it is steel, it is a permanent deformation. Looking at the other failure, buckling (in compression), does not center around the material strength, but focuses on instabilities within the geometry of the member. Like when a compression member has a lot of stress sideways, buckling will occur. Observing the equation for buckling, the failure occurs when the compressive force is more than the member's critical buckling load. In this class, (if I recall correctly from Prof. Absi), since we are dealing with pins and these pins are not too tight, the k in the equation for buckling load will be equal to 1.

Looking at the geometry of the members of our truss, we had calculated the cross-sectional areas of the members individually, where we assumed our actual truss had an Area equal to 1.0” in depth and 0.135” in thickness, and all had a different length. Using the buckling formula, we can solve $C_{cr} = \frac{\pi^2 EI}{(kL)^2}$ for each individual one.

For Young's Modulus E for A-36 steel, we can define that as 2.9×10^7 psi.

The equation for moment of inertia, and the following critical compression loads are given as (only calculated members in compression):

$$I = \frac{bh^3}{12} = \left(\frac{(.135\text{in.})^2 * (1\text{in.})}{12} \right) = 2.05 \times 10^{-4} \text{in}^4$$

$$\text{Leftmost diagonal (13.4in)} C_{cr} = \frac{\pi^2(2.9 \times 10^7 \text{ psi})(2.05 \times 10^{-4} \text{in}^4)}{(1 * 13.4)^2}$$

=326.77lbs

$$\text{Leftmost vertical (10.25in)} C_{cr} = \frac{\pi^2(2.9 \times 10^7 \text{ psi})(2.05 \times 10^{-4} \text{in}^4)}{(1 * 10.25)^2}$$

=558.48lbs

$$\text{center vertical (10.2in)} C_{cr} = \frac{\pi^2(2.9 \times 10^7 \text{ psi})(2.05 \times 10^{-4} \text{in}^4)}{(1 * 10.2)^2}$$

=563.96lbs

$$\text{centerright diagonal (13.4in)} C_{cr} = \frac{\pi^2(2.9 \times 10^7 \text{ psi})(2.05 \times 10^{-4} \text{in}^4)}{(1 * 13.4)^2}$$

=326.77lbs

$$\text{rightmost vertical (10.4in)} C_{cr} = \frac{\pi^2(2.9 \times 10^7 \text{ psi})(2.05 \times 10^{-4} \text{in}^4)}{(1 * 10.4)^2}$$

=542.48lbs

$$\text{Inside top (9.41in)} C_{cr} = \frac{\pi^2(2.9 \times 10^7 \text{ psi})(2.05 \times 10^{-4} \text{in}^4)}{(1 * 9.41)^2}$$

=662.63lbs

Length(in)	Compression Ccr(lbs)
9.41	662.63
13.4	326.77
10.25	558.48
10.2	563.96
13.4	326.77
10.4	542.48

Table 4: List of critical compression for each of our real members based on their Length L

The maximum load P that we could theoretically apply before our truss fails in buckling compression is 326.77lbs. Calculate below where the values of these buckling compression values, which only include the components that undergo compression. For the top of the steel beam, we measure from joint to joint, which was around 9.41 in each.

The maximum load P that we can apply before yielding tension is

Given the table below, we can see how the yielding was calculated and how the smaller Max yield value would determine what our maximum load is 4860 lbs.

Member:	Tensile Force	Cross-sectional area	Yield Stress(A36 steel)	Max axial Load	Max Allowed P yield
A-H	350	0.135	36,000	4860	4860
E-F	350	0.135	36,000	4860	4860
G-F	247.49	0.135	36,000	4860	6873
G-H	247.49	0.135	36,000	4860	6873

Table 5: The values needed to calculate the maximum P.

Formatted: Not Highlight

Analyzing our data, we can see that our members in compression would undergo buckling way earlier than when the other members experience yielding. Looking at the definition of yielding and buckling, we know this makes sense because yielding is based off the material and knowing that we are working with a strong material like steel, it would make sense that we have a high Max yield load.

Step 5: Comparing calculations with lab results (10pts)

Our analytical calculations did not match the test results. To first list off the differences between our actual truss versus our design, the design had members of all lengths 12in, while some of our actual trusses were as low as 10.2" and as large as 13.4". Since our measurements for those would be off, it would be sensed that our yielding and buckling did not occur at what was theoretically assumed. However, during testing, since none of our truss members yielded or buckled, it looks like our truss failed due to a welding issue instead. This is apparent, as us after a picture of our truss showed no deformation and had a broken weld instead. Because of this our calculations did NOT predict failure mode because we had other problems within our truss. Since our members were not perfectly welded together, it's safe to assume that the max load not only relied on the yield and buckling calculations but also the strength of our weld.

Step 6: Designing the Community Bridge (10pts)

The truss was analyzed under three different loading arrangements representing possible placements of landscaping area, community building, and a playground. The loads were applied at joint along the top chord, and for each configuration, the internal forces in every truss member were computed using the method of joints.

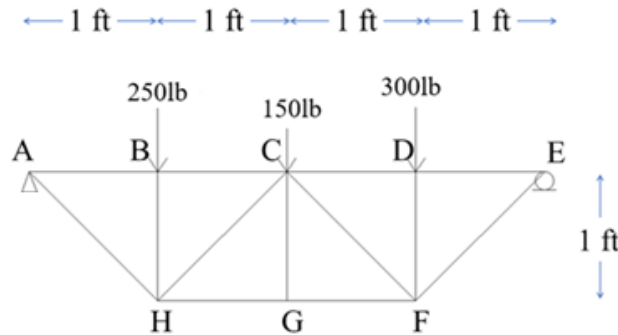


Figure 7: Example of a load combination

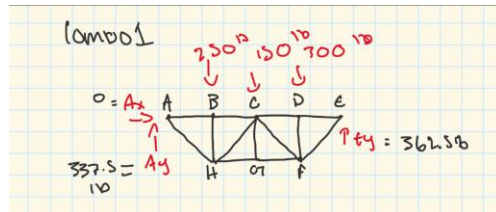


Figure 8. Global free-body diagram of the bridge truss under Load Combination 1 with 250 lb at joint B, 150 lb at joint C, and 300 lb at joint D, showing support reactions at A and E.

Member	Force (lb.)	Type
AB	-337.5	Compression
BC	-337.5	Compression
CD	-362.5	Compression
DE	-362.5	Compression

AH	477.3	Tension
BH	-250.0	Compression
HC	-123.7	Compression
HG	425	Tension
CG	0	Zero
GF	425	Tension
DF	-300.0	Compression
CF	-88.4	Compression
FE	512.7	Tension

Table 6. Member forces for Load Combination 1.

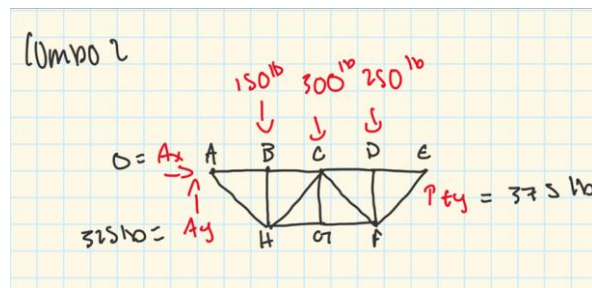


Figure 9. Global free-body diagram of the bridge diagram of the bridge truss under Load Combination 2 with 150 lb at joint B, 300 lb at joint C, and 250 lb at joint D, showing support reactions at A and E.

Member	Force (lb.)	Type
AB	-325.0	Compression
BC	-325.0	Compression
CD	-375.0	Compression
DE	-375.0	Compression
AH	459.6	Tension
BH	-150.0	Compression
HC	-247.5	Compression
HG	500	Tension
CG	0	Zero
GF	500	Tension
DF	-250.0	Compression
CF	-176.8	Compression
FE	530.3	Tension

Table 7. Member forces for Load Combination 2.

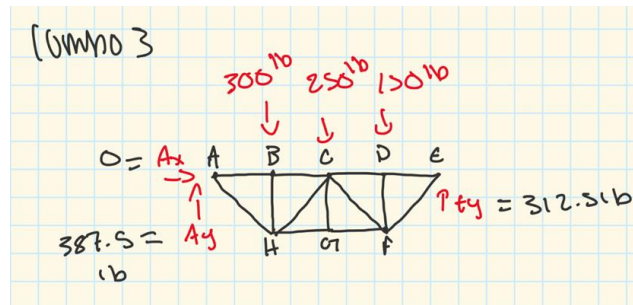


Figure 10. Global free-body diagram of the bridge diagram of the bridge truss under Load Combination 3 with 300 lb at joint B, 250 lb at joint C, and 150 lb at joint D, showing support reactions at A and E.

Member	Force (lb.)	Type
AB	-387.5	Compression
BC	-387.5	Compression
CD	-312.5	Compression
DE	-312.5	Compression
AH	548	Tension
BH	-300.0	Compression
HC	-123.7	Compression
HG	475	Tension
CG	0	Zero
GF	475	Tension
DF	-150.0	Compression
CF	-229.8	Compression
FE	441.9	Tension

Table 8. Member forces for Load Combination 3.

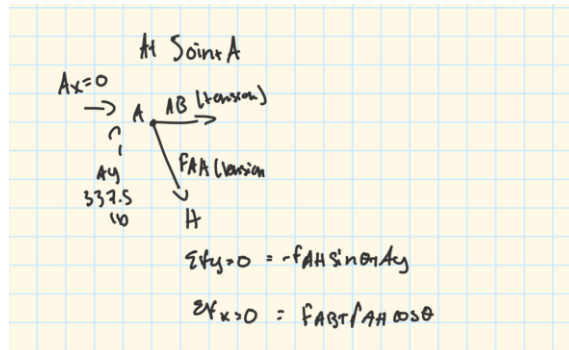


Figure 11. Joint A free-body diagram illustrating the forces in members AB and AH and the vertical support reaction at A

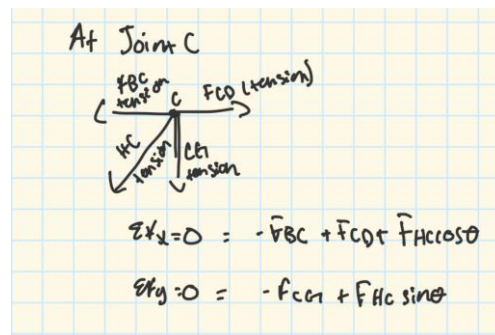


Figure 12. Joint C free-body diagram showing the axial forces in members of BC, CD, CG, and HC used in the method-of-joints analysis.

Member AH was selected for the yielding example because it experiences the largest tensile force of any member across all three combinations. Since tensile yielding is governed directly by axial stress, the member carrying the highest tension provides the most informative calculation.

$$P = 548lb \quad A = 0.135in^2$$

$$\sigma = \frac{|P|}{A}$$

$$\sigma = \frac{548}{0.135} = 4,059 \text{ psi}$$

$$4,059 \text{ psi} < 36,000 \text{ psi}$$

Member AH does not yield in tension

Member HC was selected for the buckling examples because, although its compressive force is like other thin members, its longer unsupported length reduces its Euler buckling capacity, making it the most critical member with respect to buckling.

$$P = 247.5 \text{ lb} \quad L = 13.4in \quad I = 2.05 \times 10^{-4}$$

$$P_{cr} = \frac{\pi^2 EI}{L^2} = \frac{\pi^2 (29 \times 10^6) (2.05 \times 10^{-4})}{(13.4)^2} \approx 327 \text{ lb}$$

$$247.5 \text{ lb} < 327 \text{ lb}$$

Member HC does not buckle.

Comparison of internal forces across the three load scenarios shows that Combination 1 consistently produces the smallest axial forces, while Combination 2 results in moderate forces. Combination 3 generates the largest tension and the largest compression, making it the controlling load case for design. Therefore, the community should prefer load arrangements like Combination 1, as they yield the lowest stress and highest safety margins. Anything resembling Combination 3 should be avoided for regular use because they impose the greatest demands on several members, even though the truss remains structurally safe.

Step 7: Checking for Stability on Support (5pts)

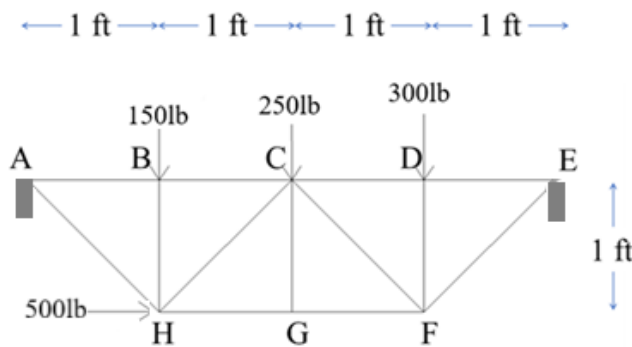


Figure 12: Truss with Horizontal Load

The friction coefficient at A is $\mu_A = 0.5$ and the friction coefficient at E is $\mu_E = 0.35$.

$\mu_A = 0.5$
 $\mu_E = 0.35$
 $\sum F_x = 0: A_x + E_x = 700 \text{ lb}$
 $\sum M_A = E_y \times 4 \text{ ft} - 300(3) - (250 \times 2 \text{ ft}) - (150 \times 1 \text{ ft}) + (500 \text{ lb} \times 1 \text{ ft}) = 0$
 $E_y = 262.5 \text{ lb}$
 $A_y = 700 - E_y = 437.5 \text{ lb}$
 $A_x = 0$
 B) $F_{A, \max} = \mu_A \cdot A_y = 218.75 \text{ lb}$
 $F_{E, \max} = \mu_E \cdot E_y = 93.75 \text{ lb}$
 $F_{\max} = 309.875 \text{ lb}$
 $F_{\max} < 500 \text{ lb}$, so the truss will slip & both slip simultaneously

Figure 14. Work shown for parts A, B, C.

The Vertical Support $A_y = 432.5\text{lb}$, and $E_y = 267.5\text{lb}$

The maximum horizontal friction forces total out to the sum of the friction forces which is 309.875lb . Since this is less than the wind (500lbs), the truss will begin to slip. But the truth is, since even in a scenario where the friction forces are maxed out, the sums of them are not greater than 500lb (the load) and therefore will cause the truss to begin slipping simultaneously. For the friction coefficients to have the truss stay in equilibrium, they would both need to be greater than 0.5 . This would mean the material is going to be rougher than concrete, which has a range of 0.3 - 0.5 . Something like steel on steel, which has a coefficient of 0.74 - 0.78 (Engineer Excel, 2023).

What would you suggest friction coefficients be for A and E to remain in static equilibrium under this load configuration? What type of material would you use for that to happen? Cite your sources.

Step 8: Conclusion and letter to the community (10pts)

Dear People of Nashville and to whomever it concerns,

We are writing to propose the construction of a cap over I-40 near Jefferson Street, a necessary step to repair a decade-old division. The interstate construction severed the historic North Nashville community, creating a lasting barrier to opportunity, health, and connectivity. A cap is the most direct engineering solution to this problem, transforming a source of division into a platform for unity. The technical benefits would immediately restore safe pedestrian and bicycle access, reconnecting neighborhoods with better economic activity. It would also create new public land for community use while also acting as a nice step towards reducing harmful noise and air pollution for adjacent residents. Our plan for developing the cap would include using a steel truss support system that would be placed over the interstate. This truss would then attempt to minimize traffic disruption, and although it would create temporary noise and construction, which would cause some lane closures and such, the impact will be much greater by bringing in communities. Our design uses this truss to support a strong enough green space and walking paths, as well as some areas for gathering. It will hopefully allow the cap to integrate nicely with the community. Our analysis, including the design and testing of a steel truss prototype, confirms that this is a viable and durable approach.

Our truss failed at a load a lot lower than the calculated buckling limit due to a poor-quality weld. This discrepancy confirms that construction quality and connection integrity are primary factors in real-world structural performance, exceeding the importance of theoretical design alone. We also learned that the Jefferson Street Cap is a practical, achievable project to finally help a community that was severely impacted by I-40.

This project has taught both me and my partner much about how engineers are quite analytical and careful when designing something that involves and impacts so many people. The prototype we made additionally gave me and my partner a lot to think about in terms of what can go wrong, and what we should start looking out for in case of errors or potential disasters, such as our faulty weld.

To end this report, the project involving our designed truss is very significant as it taught me and my partner the social importance of these underrepresented minorities and how they have gone through so much when changes in our city occur. The concepts we apply in this project allow us to fully grasp how much of an impact one's creation can cause.

Bibliography:

Federal Highway Administration. (n.d.). *Original intent: Purpose of the interstate system 1954-1956*. Retrieved December 10, 2025, from <https://highways.dot.gov/highway-history/interstate-system/original-intent-purpose-interstate-system-1954-1956>

America Walks. (2023). *Interstate construction devastated her Nashville neighborhood*. Retrieved December 10, 2025, from <https://americawalks.org/interstate-construction-devastated-her-north-nashville-neighborhood/>

Charles Gibbia. (2023, June). *Progress underway on Dallas highway cap over Interstate 35E*. The Architect's Newspaper. Retrieved December 10, 2025, from

<https://www.archpaper.com/2023/06/progress-underway-on-dallas-highway-cap-over-interstate-35e/>

PennDOT. (2025). *Cap construction update: Fall 2025*. Retrieved December 10, 2025, from <https://95revive.com/news/cap-construction-update-fall-2025/>

Nemo, L. (2023, November 13). *Dallas begins construction on cap park over highway*. Civil Engineering Online. Retrieved from <https://www.asce.org/publications-and-news/civil-engineering-source/civil-engineering-magazine/article/2023/11/dallas-begins-construction-on-cap-park-over-highway>

Hammon, M. (2023, December 15). *Opening of Denver's new freeway cap park triggers gentrification fears*. Planetizen. Retrieved from <https://www.planetizen.com/news/2023/12/126716-opening-denvers-new-freeway-cap-park-triggers-gentrification-fears>

Federal Highway Administration. (n.d.). *Belvedere Park Highway Cap Park Project*. Retrieved from https://www.fhwa.dot.gov/ipd/project_profiles/ca_belvedere_park.aspx

EngineerExcel. (2023). *Highest coefficient of friction: Applications and analysis*. Retrieved from <https://engineerexcel.com/highest-coefficient-of-friction/>